

COS2021 - Homework 3

Version: 00

Overview

This homework is designed for you to practice classification using several methods. You must pick two methods of the three mentioned below:

- Naive Bayes Classification
- K-nearest Neighbor Classification
- Decision Trees (C4.5, ID3, or C5.0)

You are welcome to try additional classification methods; however, you must try two of these.

Contents

Accompanying this PDF is a file with a tasting profile for different brands of Scotch whiskey. The file's columns are as follow:

- NAME: The name of the brand
- Columns 'wyne' to 'quick': are the attributes associated with each brand.
- Columns 'islay' to 'islands': are the categories or targets (i.e., the region of Scotland where the whiskey is produced).

The attributes are all 0 or 1 indicating that the whiskey has this property. The attributes belong to different meta-level properties, like taste. The attributes are not mutually exclusive. A whiskey can have a smokey and sweet taste.

Likewise, the categories are all 0 or 1 indicating where it came from.

The last three rows, named NEW1, NEW2, and NEW3, are the unlabeled examples. The rest of the rows are the labeled data. Every labeled example has a 1 in one of the regions. The unlabeled examples have all zeros for the categories, meaning I have removed where they come from.

Task

Your job is to use two of the three methods mentioned above to uncover the removed labels (region) for NEW1, NEW2, and NEW3. I should note that these three whiskeys are real; I did not invent them. You will not have the actual labels, so your predictions must come directly from the data. So, I should also note, I am not sure the task will produce strong results. This is part of the fun (I think).

You may use any supporting libraries or you may implement the algorithms yourself. You may do all three of the methods. The goal is not just to come up with a label but also to uncover how strong your predictions are. All three of these methods can be used to get a sense of how likely or pure the class an unknown example falls into.

I would like for you to try the predictions with all attributes. You are welcome as a follow-up to remove attributes.

Confidence in your predictions should be backed up with numbers or methodology. The confidence should come from the predictions of multiple methods. You are welcome to ignore the results of one method if you believe its results are less valuable, but you should still mention it in your write-up. You must mention the parameters you used for the different methods. For example, what was the value of K you use if you use K-nearest neighbor.

Submission

- Your python3 code. I prefer plain .py files. Not notebooks.
- The dataset itself (makes it easier for me not to have to copy it into everyone's directory).
- A professional PDF write-up of a few pages (including graphs) about your predictions and how confident you are (not word, pages or some other format).